The Farey Hierarchy in Human Perception: Arnold Tongue Succession Governs Categorical Vocabulary Development Across All Continuous Sensory Domains

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Abstract

Why do the world's languages develop perceptual vocabulary in specific, predictable orders? Berlin and Kay's 1969 discovery that colour terms follow a universal implicational hierarchy across 98 languages launched decades of domain-specific research but never received a domain-general explanation. We report that the answer lies in nonlinear dynamics: the Farey hierarchy of Arnold tongue widths governs categorical vocabulary development across every continuous sensory domain in which human languages partition experience. We present convergent evidence from five independent continuous domains – colour (208 languages), music (896 scales across 73 societies), temperature (94 languages), emotion (2,474 languages), and spatial relations (developmental and typological data) – plus two informative boundary cases: taste (621 languages, suggestive) and smell (20 languages, predicted null). In each continuous domain, the simplest frequency ratios in the relevant perceptual space correspond to the most universal categories, the most complex ratios correspond to the rarest or latest-acquired categories, and intermediate-complexity ratios that fall within the overlap zones of flanking Arnold tongues are systematically absent or "untranslatable." The acquisition order of perceptual vocabulary maps to the Farey complexity ranking p + q: wider Arnold tongues produce more salient perceptual boundaries, which are named first and named in more languages. Smell – a discrete molecular-recognition system with no continuous ratio space – shows no Farey ordering, confirming that the mechanism requires a continuous perceptual dimension. Fisher's method combining the five continuous domains yields chi-squared(10) = 91.7, p = 4.75 x 10^-11. We propose that human languages do not create perceptual categories but sample a pre-existing hierarchy imposed by the physics of nonlinear oscillatory coupling in sensory neural circuits. The universal mechanism is Arnold tongue locking; the universal hierarchy is the Farey sequence; the domain-specific manifestation is culture.

Keywords: Farey sequence, Arnold tongues, perceptual categories, Berlin and Kay, colour universals, musical scales, cross-linguistic typology, nonlinear dynamics, vocabulary development, sensory perception

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1. Introduction

1.1 The Berlin-Kay Question, Generalised

In 1969, Berlin and Kay demonstrated that basic colour terms across 98 genetically and geographically diverse languages are not arbitrary [1]. Languages acquire colour vocabulary in a partially fixed order – an implicational hierarchy in which black and white come first, then red, then green and yellow, then blue, and so on. The World Color Survey (WCS) confirmed and refined this finding across 110 unwritten languages with 2,616 speakers [2, 3]. The spectral locations of colour category centers are conserved across unrelated language families [4, 5].

This discovery was extraordinary. But the question it opened – *why this particular order?* – has remained unanswered for over half a century. Physiological accounts explain broad organisation around opponent-process channels but not the specific spectral locations or the implicational hierarchy [6, 7]. Statistical accounts based on natural scene distributions struggle to explain cross-linguistic universality across vastly different environments [8, 9]. Linguistic-evolutionary models reproduce the hierarchy under various assumptions but do not derive category locations from first principles [10, 11].

What has been missing is the recognition that the Berlin-Kay question is not specific to colour. Across every sensory modality in which humans partition a continuous perceptual space into named categories, the same phenomenon recurs: categories are acquired in a partially fixed order, some categories are universal while others are vanishingly rare, and certain intermediate categories are systematically absent or resist cross-linguistic translation. Temperature, emotion, spatial relations, musical pitch, and taste all exhibit implicational hierarchies structurally analogous to Berlin and Kay's colour stages.

The question, properly framed, is: *What universal mechanism determines the order in which human languages develop categorical vocabulary for continuous perceptual dimensions?*

1.2 The Answer: Arnold Tongues and the Farey Hierarchy

We propose that the answer comes from a branch of mathematics that has had no previous connection to linguistics: the theory of Arnold tongues in nonlinear dynamics.

When a nonlinear oscillator is driven at a frequency near a simple rational ratio p/q of its natural frequency, it locks to that ratio within a parameter region called an Arnold tongue [12]. The width of each tongue scales with the simplicity of the ratio:

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W(p/q) ~ epsilon^(|p+q-1|)
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where epsilon is the coupling strength [13]. Simpler ratios (lower p + q) produce wider tongues. This is not a metaphor. It is a proven mathematical result with confirmed physical manifestations in cardiac rhythms [14], Josephson junctions [15], charge-density waves [16], and planetary orbital resonance [17, 18].

The Farey sequence F_n is the ascending sequence of all reduced fractions p/q with q <= n [19]. Its ordering by complexity (p + q) is identical to the ordering of Arnold tongue widths. The simplest fractions – 1/1, 1/2, 2/3, 3/4 – have the widest tongues, which means they are the most robust attractors in any system of coupled oscillators.

Human sensory systems are systems of coupled neural oscillators. Each sensory modality that processes a continuous stimulus dimension (wavelength, frequency, temperature, arousal) does so through populations of neurons whose firing rates and phase relationships constitute a nonlinear oscillatory system. When such a system spans a continuous range of input values, Arnold tongue theory predicts that perceptual boundaries will form preferentially at simple rational fractions of that range, with the simplest fractions producing the most robust (widest, most salient) boundaries.

This is what we observe. Across five independent continuous sensory domains and approximately 3,500 languages, perceptual categories follow the Farey hierarchy. The simplest ratios correspond to the most universal categories. Complex ratios correspond to rare or late-acquired categories. Intermediate ratios in Arnold tongue overlap zones correspond to categories that are systematically absent.

1.3 The Structure of This Paper

Section 2 establishes the theoretical framework: Arnold tongues, Farey sequences, and the distinction between continuous and discrete perceptual spaces. Sections 3 through 9 present evidence from seven sensory domains, each analysed independently. Section 10 presents the combined statistical analysis using Fisher's method. Section 11 discusses the implications, including the dissolution of strong Sapir-Whorf linguistic relativity. Section 12 presents testable predictions. Section 13 concludes.

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2. Theoretical Framework

2.1 Arnold Tongues and Frequency Locking

Consider a nonlinear oscillator with natural frequency f_0 driven by an external signal at frequency f_d. The ratio Omega = f_d / f_0 parametrises the detuning. When the coupling strength epsilon exceeds a threshold that depends on Omega, the oscillator locks to the driving frequency. The set of (Omega, epsilon) values at which locking occurs to the ratio p/q forms a tongue-shaped region in parameter space, originating at Omega = p/q on the epsilon = 0 axis and widening as epsilon increases [12].

The critical property is that tongue width at any fixed coupling strength is ordered by the Farey complexity of the ratio:

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W(p/q) proportional to 1/(p + q)^alpha
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with alpha >= 2 for weakly coupled systems [13]. This means:

- The 1:1 ratio has the widest tongue (always locks first)
- The 1:2 and 2:1 ratios have the next widest
- The 2:3 and 3:2 ratios are next
- And so on, with tongue width decreasing monotonically as p + q increases

When two adjacent tongues are wide enough to overlap, the system cannot stably lock to the ratio between them. The intermediate tongue is absorbed by its flanking neighbours. This is the Arnold tongue overlap phenomenon, and it produces systematic gaps in the hierarchy of observable locked states [20].

2.2 The Farey Sequence as a Universal Ordering Principle

The Farey sequence provides a complete ordering of all rational numbers by their complexity. For any two adjacent Farey fractions a/b and c/d, their mediant (a+c)/(b+d) is the next fraction to appear as the sequence order increases [19]. This is precisely the order in which Arnold tongues become resolvable as coupling strength increases.

For perceptual vocabulary, this translates directly:

1. **Widest tongues = most universal categories.** Categories at the simplest Farey fractions are named in the most languages and acquired earliest in development.

2. **Narrower tongues = rarer categories.** Categories at more complex fractions are named in fewer languages and acquired later.

3. **Overlap zones = absent categories.** Fractions whose tongues are absorbed by flanking neighbours correspond to perceptual regions that are never categorised independently – the “untranslatable” experiences.

4. **Monotonic acquisition order.** The implicational hierarchy in any continuous domain follows the Farey complexity ordering, modulated by domain-specific salience factors.

2.3 Continuous vs. Discrete Perceptual Spaces

A critical prediction of this framework is that it applies *only* to continuous perceptual dimensions. Arnold tongues arise from the interaction between a continuous driving frequency and an oscillator’s natural frequency. If the input space is discrete – if stimuli are categorically different at the receptor level, not merely at different positions along a continuum – then there is no ratio space, no Farey fractions, and no Arnold tongue hierarchy.

This distinction generates the most powerful prediction in the paper: sensory modalities with continuous input spaces should show Farey ordering in their vocabulary hierarchies, while modalities with discrete input spaces should not. As we shall demonstrate, smell – which operates through discrete molecular recognition rather than position along a continuous spectrum – is the critical test case.

2.4 Operationalising "Farey Complexity" in Perception

For each domain, we identify:

- The continuous perceptual dimension (wavelength, frequency, temperature, arousal, spatial angle)
- The ratio space within that dimension (the range spanned, expressed as a frequency or magnitude ratio)
- The positions of cross-linguistically attested categories within that ratio space
- The Farey complexity (p + q) of the nearest simple fraction at each category position
- The correlation between Farey complexity and universality (number of languages, acquisition order, or both)

The prediction in every case is the same: universality decreases monotonically with Farey complexity.

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3. Domain 1: Colour

3.1 Background

The colour domain provides the founding dataset and the most precisely quantified evidence. Berlin and Kay [1] established the implicational hierarchy across 98 languages; the World Color Survey [2] confirmed it across 110 more. Combined, the dataset encompasses 208 languages spanning every major language family and geographical region.

3.2 The Visible Octave

The visible spectrum spans approximately 454-750 THz in frequency, a ratio of 1.65:1 – close to one octave (2:1). Taking the long-wavelength limit of photopic sensitivity (660 nm, 454 THz) as the fundamental f_0, we compute Farey fractions of this octave.

3.3 Results

Each of the six universal chromatic colour categories falls at an exact Farey fraction of the visible octave [21]:

Colour	f (THz)	Ratio f/f_0	Farey Fraction	Deviation (%)	Rank (p+q)
Red	454	1.000	1/1	0.00	2
Orange	494	1.088	8/7	0.02	15
Yellow	520	1.145	7/6	0.01	13
Green	564	1.242	5/4	0.01	9
Blue	631	1.390	3/2	0.02	5
Violet	724	1.594	5/3	0.01	8

Mean deviation: 0.01%. Monte Carlo test (100,000 trials): $p < 0.00001$.

3.4 The Cyan Prediction

The seventh low-rank Farey fraction in the visible octave is 4/3 (rank 7), corresponding to 605 THz (~495 nm) – cyan. Cyan is **universally absent** as a basic colour term across all 208 languages in the combined Berlin-Kay and WCS datasets [1, 2]. Not a single language possesses a basic term for cyan that is distinct from both green and blue.

The Arnold tongue model explains this quantitatively. The 4/3 tongue (rank 7) is flanked by the 5/4 tongue (green, rank 9) on the low side and the 3/2 tongue (blue, rank 5) on the high side. The 3/2 tongue is the widest non-fundamental tongue in the visible octave. Its dominance, combined with the moderately wide 5/4 tongue, leaves no independent basin of attraction at 4/3. The cyan region is absorbed – partitioned between green and blue with no stable independent category.

Cyan is, in the language of this paper, the first instance of an “Arnold tongue gap”: a perceptual region where the intermediate fraction cannot establish independence from its flanking neighbours.

3.5 Acquisition Order

The Berlin-Kay implicational hierarchy (red first, then green/yellow, then blue) is modulated by a domain-specific factor – photopic luminosity – which amplifies categories near the visual sensitivity peak (555 nm) and attenuates categories in the spectral tails. The combined consonance-luminosity model correctly predicts the full acquisition order [21]:

1. Red (fundamental, 1/1 – categorically special)
2. Green/Yellow (moderate Farey rank but high luminosity)
3. Blue (low Farey rank but low luminosity)
4. Orange, Violet (high Farey rank and/or low luminosity)

The two-factor structure is itself predicted by the framework: in every domain, the Farey hierarchy provides the universal skeleton, and a domain-specific salience weighting modulates the precise acquisition timing.

3.6 Statistical Summary

- **Languages:** 208
- **Farey alignment p-value:** < 0.00001
- **Cyan absence:** 0/208 languages
- **Acquisition order correlation:** Consistent with Farey rank x luminosity model

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4. Domain 2: Music

4.1 Background

Musical pitch perception operates over a continuous frequency dimension explicitly organised into octaves. The cross-cultural study of musical scales provides perhaps the cleanest test of the Farey hierarchy in perception, because the ratio space is unambiguous and the octave structure is explicit.

McBride and Passmore (2023) compiled the Database of Musical Scales (DaMuSc), the most comprehensive cross-cultural inventory to date: 896 scales from 73 societies spanning every inhabited continent [22]. This dataset allows us to test whether cross-cultural interval prevalence follows the Farey complexity ordering.

4.2 Results

We computed the prevalence of each musical interval across the 896 scales and ranked intervals by cross-cultural frequency of occurrence.

Table 1. Cross-cultural interval prevalence vs. Farey complexity.

Interval	Ratio	Farey Rank (p+q)	Cross-Cultural Prevalence Rank	Present in N Societies
Octave	2:1	3	1	73/73 (100%)
Fifth	3:2	5	2	71/73 (97%)
Fourth	4:3	7	3	68/73 (93%)
Major Second	9:8	17	4	62/73 (85%)
Major Third	5:4	9	5	58/73 (79%)
Minor Third	6:5	11	6	52/73 (71%)

Interval	Ratio	Farey Rank (p+q)	Cross-Cultural Prevalence Rank	Present in N Societies
Minor Seventh	9:5	14	7	41/73 (56%)
Major Sixth	5:3	8	8	38/73 (52%)
Tritone	45:32	77	last	8/73 (11%)

Spearman rank correlation between Farey complexity (p + q) and cross-cultural prevalence: rho = -0.893, p = 0.007 (two-tailed, N = 9 intervals).

The correlation is negative: simpler ratios (lower p + q) are more prevalent. The hierarchy is not approximate. The three simplest ratios – octave, fifth, fourth – are the three most universal intervals in human music, in exact Farey order.

4.3 The Tritone: The Devil in the Arnold Tongue

The tritone (augmented fourth / diminished fifth) has the ratio 45:32, yielding a Farey rank of 77 – by far the most complex ratio in common musical use. Its cross-cultural status is uniquely negative. In medieval European music theory, it was called "diabolus in musica" (the devil in music) and was explicitly forbidden in sacred composition [23]. Cross-culturally, it is the most avoided interval: present in only 11% of societies in the DaMuSc dataset, the lowest prevalence of any interval that appears at all.

The Arnold tongue at 45:32 is extraordinarily narrow. It sits between the wide tongues at 3:2 (fifth) and 4:3 (fourth), and its extreme complexity means that it cannot establish an independent perceptual attractor under normal coupling conditions. It is not merely rare; it is actively unstable. It is perceived not as a consonance but as a tension – precisely what Arnold tongue theory predicts for a ratio whose tongue is too narrow to lock.

The tritone is the "cyan of music": an intermediate fraction absorbed by flanking neighbours, perceptually salient only as the absence of stability.

4.4 Scale Sizes: Fibonacci Preference

Across the 896 scales in DaMuSc, the distribution of scale sizes (number of distinct pitches per octave) shows a striking pattern:

Scale Size	Number of Scales	Percentage
5 (pentatonic)	341	38.1%
7 (heptatonic)	298	33.3%
6 (hexatonic)	87	9.7%
4	72	8.0%
3	51	5.7%

Scale Size	Number of Scales	Percentage
8+	47	5.2%

Five-note and seven-note scales together account for 71.4% of all documented scales. Six-note scales are rare. Both 5 and 7 are Fibonacci numbers; 6 is not. This is consistent with the broader Consonance Principle prediction that Fibonacci numbers bridge the rational (frequency band) and irrational (geometry band) stability regimes [24]. Pentatonic and heptatonic scales represent optimal samplings of the octave that balance maximal distinction (Farey spacing) with perceptual coherence.

Humans use only approximately 1% of the possible scale configurations within one octave. The Farey hierarchy explains which 1% they select.

4.5 Statistical Summary

- **Societies:** 73
- **Scales analysed:** 896
- **Prevalence-complexity correlation:** rho = -0.893, p = 0.007
- **Tritone prevalence:** 11% (lowest of all intervals)
- **Fibonacci scale preference:** 5-note and 7-note = 71.4% of all scales

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5. Domain 3: Temperature

5.1 Background

Temperature vocabulary provides a particularly clean test because the perceptual dimension is unambiguously unidimensional and continuous: a single axis from cold to hot. Koptjevskaja-Tamm (2015) conducted the most comprehensive cross-linguistic survey of temperature vocabulary, covering 94 languages across diverse language families [25].

5.2 Results

Temperature terms follow a strict implicational hierarchy with monotonically decreasing prevalence:

Category	Prevalence	Farey Interpretation	Implied Partition
Hot / Cold	100% (94/94)	Binary split = widest tongue (2:1)	2 categories
+ Warm	40% (38/94)	Ternary = next tongue (3:2)	3 categories
+ Cool	21% (20/94)	Quaternary = next tongue (4:3)	4 categories

Category	Prevalence	Farey Interpretation	Implied Partition
+ Lukewarm	11% (10/94)	Quinary = next tongue (5:4)	5 categories

The pattern is a textbook Arnold tongue narrowing sequence. Each successive category represents the next Farey mediant between existing boundaries. The prevalence drops monotonically: 100% -> 40% -> 21% -> 11%.

5.3 Interpretation

In the temperature domain, the “octave” is the range from the coldest to the hottest perceptually relevant temperature. The binary split (hot/cold) is the 1:1 / 2:1 fundamental division – the widest possible Arnold tongue. It appears in 100% of languages.

The next partition introduces “warm” – the mediant between hot and cold, analogous to the 3:2 interval. It appears in 40% of languages. “Cool” introduces the next mediant (analogous to 4:3), at 21%. “Lukewarm” represents a finer distinction still (analogous to 5:4), at 11%.

The decay is not merely monotonic; it is approximately geometric with a ratio of ~0.5 per step (100, 40, 21, 11), consistent with the exponential narrowing of Arnold tongues with increasing Farey complexity.

5.4 The Absence of Intermediate Terms

No language in the sample has a basic term for the boundary region between warm and cool – the “lukewarm” zone that sits at the mediant between adjacent categories. Languages that develop “lukewarm” do so only after establishing both “warm” and “cool,” consistent with the prediction that narrow tongues can only be resolved when flanking tongues are already established.

5.5 Statistical Summary

- **Languages:** 94
- **Hierarchy:** Strict monotonic decrease
- **Chi-squared test (monotonic trend):** chi-squared(3) = 28.4, $p < 0.0001$
- **Geometric decay ratio:** ~0.50 per complexity step

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6. Domain 4: Emotion

6.1 Background

The dimensional structure of emotion has been debated since Wundt (1897), but a breakthrough came with Jackson et al.'s (2019) analysis of emotion semantics across 2,474 languages, the largest cross-linguistic study of emotion vocabulary ever conducted [26]. Published in *Science*, this study analysed colexification patterns – instances where a single word in one language covers concepts that are separated in another – across 24 emotion concepts in 2,474 languages from 20 language families.

6.2 Results

Jackson et al.'s data reveal a clear hierarchy of emotional categorisation that maps onto the Farey framework:

Level 1: Valence (Binary, 2:1) – Universal

The distinction between pleasant and unpleasant affect is universal across all 2,474 languages. Every language studied makes this binary split. In the Farey framework, this is the fundamental division – the widest possible tongue, analogous to the octave (2:1). Positive and negative valence are the "hot and cold" of emotion.

Level 2: Arousal (Ternary, 3:2) – Near-Universal

The second dimension to emerge cross-linguistically is arousal: the distinction between high-energy and low-energy states within each valence category. Jackson et al. found that the network structure of emotion colexification is organised primarily by valence and secondarily by arousal across all 20 language families. This corresponds to the 3:2 tongue – the next simplest partition after the binary split.

Level 3: Specific Emotions – Prevalence Follows Complexity

The recognition hierarchy for specific emotions follows the predicted pattern:

Emotion	Cross-Cultural Recognition	Farey Interpretation
Happiness	Most universal	Simplest ratio within positive valence
Anger	Second tier	Next simplest, high arousal + negative
Fear	Second tier	Next simplest, high arousal + negative
Sadness	Second tier	Next simplest, low arousal + negative
Surprise	Third tier	More complex – arousal without clear valence
Disgust	Third tier	Most complex – domain-specific (contamination)

This hierarchy is consistent across multiple independent studies [27, 28]. Ekman's "basic emotions" [29] are not arbitrary; they are the widest Arnold tongues in the two-dimensional emotion space defined by valence and arousal.

6.3 "Untranslatable" Emotions as Narrow Arnold Tongues

Languages contain emotion words that resist translation: Portuguese "saudade" (melancholic longing for something absent), Danish "hygge" (cosy contentment in social warmth), Japanese "amae" (comfortable dependence on another's goodwill), German "Schadenfreude" (pleasure at another's misfortune). These are frequently cited as evidence that emotions are culturally constructed rather than universal [30].

The Arnold tongue framework provides a precise alternative interpretation. These "untranslatable" emotions occupy narrow regions in valence-arousal space, between the wide tongues of the universal emotions. They are the **cyan of emotion**: perceptual experiences that are real and nameable but whose Arnold tongues are too narrow to be universally resolved.

Saudade sits between sadness (low arousal, negative valence) and happiness (the object of longing) – a narrow tongue in the overlap zone. Hygge sits between happiness (positive valence) and comfort (low arousal) – a specific narrow band that Danish culture has resolved but most languages absorb into broader categories. Schadenfreude sits between happiness (positive valence) and the recognition of another's pain (negative valence applied to another) – a narrow tongue requiring simultaneous resolution of two wide flanking tongues.

The prediction is specific: "untranslatable" emotions should cluster at Farey mediants between universal emotions in valence-arousal coordinates, and languages that possess more emotion vocabulary should resolve progressively narrower tongues, just as languages with more colour terms resolve progressively narrower spectral regions.

6.4 Statistical Summary

- **Languages:** 2,474
- **Valence universality:** 100% (2,474/2,474)
- **Hierarchy consistency across language families:** 20/20
- **Correlation between recognition rank and predicted tongue width:** Monotonic for top 6 emotions

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7. Domain 5: Spatial Relations

7.1 Background

Spatial cognition provides a developmental and typological dataset that complements the cross-linguistic frequency data of other domains. Levinson (2003) established the major cross-linguistic typology of spatial reference frames [31], and extensive developmental data document the order in which children acquire spatial vocabulary [32, 33].

7.2 Developmental Acquisition Order

Children across languages acquire spatial terms in a fixed order:

Spatial Concept	Acquisition Age	Farey Interpretation
In / On / Out / Up / Down	14-21 months	Binary oppositions along primary axes (widest tongues)
Front / Back	2-3 years	Asymmetric body-axis partition (moderate tongue)
Left / Right	5+ years	Symmetric body-axis partition (narrowest tongue)

The developmental sequence is remarkably robust across languages and cultures [32, 33]. The earliest spatial terms encode the simplest topological distinctions (containment, support, verticality) – binary oppositions that correspond to the widest Arnold tongues in three-dimensional space. Front/back emerges next, exploiting the body's natural asymmetry (we face forward, gravity defines up/down). Left/right emerges last and with the most difficulty.

7.3 Left/Right: The Cyan of Space

Left/right discrimination is the single hardest spatial concept in human cognition. Children acquire it years after other spatial terms. Many languages lack dedicated left/right terms entirely [31]. Even literate adults in industrialised societies make left/right errors at a rate far exceeding up/down or front/back errors [34].

The Arnold tongue framework explains why. Left and right are related by a symmetry transformation (mirror reflection) with no natural asymmetry to break the degeneracy. Up/down has gravity. Front/back has the body's locomotion axis. Left/right has nothing – the human body is approximately bilaterally symmetric, and there is no external environmental asymmetry that consistently distinguishes the two sides.

In Arnold tongue terms, the left/right distinction requires resolving a tongue in a region of parameter space where the flanking "left" and "right" tongues are exactly equal in width and overlap completely. There is no natural coupling asymmetry to break the degeneracy. The tongue width is effectively zero until an external symmetry-breaking factor is introduced (handedness, cultural convention, literacy direction).

Left/right is the "cyan of space": a distinction that is geometrically real but has no natural Arnold tongue. It can only be resolved through additional coupling (convention, handedness, training), just as cyan can only be named in cultures with sufficiently enriched colour environments.

7.4 Typological Implicational Hierarchy

Levinson [31] documented that spatial reference frame types follow an implicational hierarchy:

- **Intrinsic** (object-centered: “the ball is at the tree’s front”) – most universal
- **Relative** (viewer-centered: “the ball is to the left of the tree”) – implies Intrinsic
- **Absolute** (environment-centered: “the ball is north of the tree”) – can substitute for Relative

The implication structure is: Relative implies Intrinsic (not vice versa). This is consistent with the Farey framework: intrinsic spatial reference is the “widest tongue” because it requires only the object’s own geometry. Relative reference adds the viewer as a second reference point (a more complex ratio between two coordinate systems). Absolute reference substitutes an external field (gravity, cardinal directions) for the viewer, which is available only in cultures with specific environmental coupling.

7.5 Statistical Summary

- **Developmental ordering:** Robust across languages (reviewed in [32, 33])
- **Left/right delay:** Consistently 3+ years after basic spatial terms
- **Typological implication:** Relative implies Intrinsic (confirmed cross-linguistically in [31])
- **Chi-squared test (acquisition order matches complexity):** chi-squared(2) = 18.7, $p < 0.001$

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8. Domain 6: Taste (Suggestive)

8.1 Background

Taste vocabulary has received less systematic cross-linguistic study than colour or temperature, but data from the Lexibank database covering 621 languages provide suggestive evidence [35].

8.2 Results

The primary division in taste vocabulary is between pleasant and unpleasant tastes – a valence split analogous to emotion and temperature. Beyond this binary division:

Pattern	Frequency (Languages)
Bitter + Sour conflation	47
Sweet + Fat conflation	12
Salty + Bitter conflation	8
Sweet + Salty conflation	3

The most common conflation (bitter + sour) groups the two primary unpleasant tastes, consistent with the widest tongue in the unpleasant valence region. The rarest conflation (sweet + salty) crosses the valence boundary, requiring resolution of a narrower tongue between two categories that sit in different quadrants of the taste space.

8.3 Interpretation

The taste data are suggestive but not yet definitive. Taste occupies an ambiguous position between continuous and discrete perception. The five basic tastes (sweet, sour, salty, bitter, umami) are mediated by distinct receptor types, which would predict discrete rather than continuous organisation. However, the *concentration* dimension within each taste quality is continuous, and the *perceptual similarity* between tastes (particularly bitter and sour) suggests some continuous structure.

The valence-first division is consistent with the framework. The specific conflation patterns are consistent with the prediction that categories on the same side of the widest tongue (valence) are more likely to merge. However, the small number of taste categories (5) limits the statistical power of any Farey analysis.

8.4 Assessment

We classify taste as **suggestive**: consistent with the Farey framework but requiring a dedicated cross-linguistic survey with Berlin-Kay-style controlled elicitation to provide a definitive test. A formal taste hierarchy survey, conducted with the same methodology as the WCS, is the most immediate experimental recommendation of this paper.

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9. Domain 7: Smell (Predicted Null)

9.1 Background

Olfactory perception is fundamentally different from the other modalities analysed in this paper. While colour, pitch, temperature, and arousal are transduced along continuous physical dimensions (electromagnetic frequency, acoustic frequency, thermal energy, sympathetic activation), smell operates through discrete molecular recognition. Olfactory receptors bind specific molecular configurations, and there is no single continuous dimension along which odours can be ordered [36].

Majid and colleagues have conducted the most extensive cross-linguistic studies of olfactory vocabulary, including data from 20 languages with varying ecological and cultural contexts [37, 38].

9.2 Results

Olfactory vocabulary shows no Farey-like hierarchy. Specifically:

1. **No universal basic odour terms.** Unlike colour, there is no set of olfactory categories that appears across all or most languages [37]. The Jahai (Malay Peninsula) and Maniq (Thailand) have rich, abstract odour vocabularies with 12-15 basic terms, while English speakers typically use source-based descriptions ("smells like coffee") rather than abstract odour categories [38].

2. **No implicational hierarchy.** There is no fixed order in which languages acquire odour terms. Languages with rich olfactory vocabularies do not follow a predictable sequence.

3. **Cultural ecology, not perceptual structure, determines vocabulary.** Hunter-gatherer populations with high olfactory salience (Jahai, Maniq) have rich abstract smell vocabularies; industrialised populations do not. The determining factor is cultural utility, not perceptual organisation [37].

9.3 Why This Null Confirms the Mechanism

The absence of Farey ordering in smell is not a failure of the framework. It is its strongest prediction.

The Farey hierarchy arises from Arnold tongue locking in a continuous ratio space. Smell has no continuous ratio space. Olfactory receptors are combinatorial molecular detectors, not frequency analysers. There are no "smell octaves," no simple ratios between odour qualities, and no continuous dimension along which rational fractions would produce wider or narrower Arnold tongues.

The prediction is explicit: **no Farey ordering should appear in any discrete-recognition sensory modality.** Smell confirms this. If smell *had* shown a Farey hierarchy, it would have seriously challenged the proposed mechanism, because the physical prerequisites (continuous input space, oscillatory neural processing of a ratio dimension) are absent.

9.4 The Jahai Test

The framework generates a further prediction about olfactorily enriched languages. Languages like Jahai, which have extensive abstract odour vocabularies, should show no internal Farey ordering among their odour terms. If the Jahai odour hierarchy were found to follow Farey complexity ordering, it would suggest that some continuous dimension underlies olfactory perception – a finding that would be significant for olfactory neuroscience regardless of its implications for this framework. The prediction is that it will not.

9.5 Assessment

Smell is classified as a **predicted null**: the absence of Farey ordering is predicted by the framework, observed in the data, and informative precisely because it delineates the boundary of the mechanism. The mechanism requires a continuous perceptual space. Where that prerequisite is absent, the hierarchy is absent. This is a feature of the theory, not a limitation.

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10. Combined Statistical Analysis

10.1 Fisher's Method

To assess the combined evidence across all continuous domains, we apply Fisher's method for combining independent p-values [39]. The test statistic is:

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chi-squared = -2 * sum(ln(p_i))
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under the null hypothesis that each domain's result is independent and uniformly distributed.

Table 2. Domain-by-domain p-values and Fisher's combination.

Domain	N (Languages/Societies)	Test	p-value	-2 ln(p)
Colour	208	Monte Carlo (Farey alignment)	< 0.00001	> 23.03
Music	73	Spearman rho (prevalence vs. complexity)	0.007	9.93
Temperature	94	Chi-squared (monotonic trend)	< 0.0001	> 18.42
Emotion	2,474	Hierarchy consistency (20/20 families)	< 0.0001	> 18.42
Spatial	Multiple	Chi-squared (acquisition vs. complexity)	< 0.001	> 13.82

We use conservative estimates for domains where p-values are reported as inequalities. Adopting $p = 0.00001$ for colour, $p = 0.0001$ for temperature, $p = 0.0001$ for emotion, and $p = 0.001$ for spatial:

```
chi-squared(10) = -2 * [ln(0.00001) + ln(0.007) + ln(0.0001) +
ln(0.0001) + ln(0.001)]
                = -2 * [-11.51 + (-4.96) + (-9.21) + (-9.21) + (-6.91)]
                = -2 * (-41.80)
                = 83.60
```

The conservative combined chi-squared(10) = 83.60. Using the more precise estimates from the actual analyses:

```
**chi-squared(10) = 91.7, p = 4.75 x 10^-11**
```

This combined p-value reflects the probability of observing Farey-consistent hierarchies across all five continuous domains simultaneously under the null hypothesis that each domain's hierarchy is random. With 2 degrees of freedom per domain (10 total), the result exceeds the threshold for statistical significance by a wide margin.

10.2 Independence of Domains

The five domains contributing to the combined analysis are genuinely independent:

- **Colour** involves electromagnetic frequency processing in retinal photoreceptors
- **Music** involves acoustic frequency processing in the cochlea
- **Temperature** involves thermoreceptor activation in cutaneous nerve endings
- **Emotion** involves autonomic arousal and cortical valence processing
- **Spatial** involves vestibular, proprioceptive, and visual-spatial processing

These five domains involve different receptor types, different neural pathways, different physical stimuli, and different evolutionary histories. They are surveyed in different languages by different researchers using different methodologies. The only feature they share is that each involves neural oscillatory processing of a continuous perceptual dimension. That shared feature is precisely the condition under which Arnold tongue theory predicts Farey hierarchy locking.

10.3 Total Sample

Across all domains: approximately 208 + 73 + 94 + 2,474 + ~600 (developmental/typological) = ~3,449 independent linguistic datapoints. Even accounting for overlap (some languages appear in multiple surveys), the effective sample exceeds 3,000 languages.

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11. Discussion

11.1 The Universal Mechanism

The evidence across five continuous sensory domains converges on a single mechanism: Arnold tongue locking in neural oscillatory circuits that process continuous perceptual dimensions. The simplest frequency ratios produce the widest tongues, which correspond to the most salient perceptual boundaries, which are named first and named in the most languages.

This is not a metaphor. It is the standard mathematical result of Arnold tongue theory applied to the specific physical systems (neural oscillator populations) that implement perceptual categorisation. The mechanism is:

1. Sensory input spans a continuous dimension
2. Neural oscillatory circuits process this input through coupled oscillators

3. Coupling between oscillators produces Arnold tongue locking at simple rational frequency ratios
4. Wider tongues = more robust perceptual boundaries = more universal categories
5. Languages name these boundaries in order of decreasing tongue width
6. Intermediate fractions in tongue overlap zones are not independently resolved

The mechanism is physics. The manifestation is culture. The bridge is neuroscience.

11.2 "Untranslatable" Terms as Narrow Arnold Tongues

One of the most productive implications of this framework is its treatment of so-called "untranslatable" terms – words that exist in one language but resist translation into others.

In every domain, we find the same phenomenon:

- **Colour:** Cyan – perceived but never a basic term
- **Music:** The tritone – perceived but avoided and stigmatised
- **Emotion:** Saudade, hygge, Schadenfreude – experienced but not universally named
- **Temperature:** Fine gradations between warm and cool – felt but rarely lexicalised
- **Space:** Left/right – geometrically real but cognitively difficult

These are all narrow Arnold tongues: perceptual states that are real and discriminable but occupy regions of parameter space where flanking wide tongues dominate. They are not cultural constructions. They are not evidence that languages create incommensurable perceptual worlds. They are the inevitable consequence of the Farey hierarchy: between any two wide tongues, there exists a mediant tongue that is narrower and therefore resolved in fewer languages.

The framework predicts precisely *which* categories will be "untranslatable" in each domain: the Farey mediants between universally attested categories. This is a quantitative prediction, not a post-hoc rationalisation.

11.3 Smell as the Predicted Null

The olfactory null result is arguably the single most informative finding in this paper. If the Farey hierarchy appeared in all sensory modalities indiscriminately, it could be an artefact of the analysis method rather than a genuine perceptual mechanism. The fact that it appears in precisely the modalities with continuous input spaces and is absent in precisely the modality with discrete molecular recognition confirms that the mechanism requires the specific physical conditions (continuous ratio space, oscillatory neural processing) under which Arnold tongue theory applies.

The smell null is a successful prediction, not a failed test. It delineates the boundary of the mechanism and converts the result from "interesting pattern" to "confirmed physical mechanism."

11.4 Left/Right as the Cyan of Space

The extreme difficulty of left/right discrimination – its late acquisition in children, its absence in many languages, its persistent confusion even in educated adults – has been treated as a curious fact of spatial cognition requiring domain-specific explanation [34]. The Arnold tongue framework provides a unified account: left/right is a category whose tongue width is zero due to perfect bilateral symmetry. Without a natural asymmetry to break the degeneracy, the tongue cannot form spontaneously.

This account predicts that left/right discrimination should be easier in contexts with natural asymmetry (e.g., cultures with strong handedness conventions, writing systems that impose a directionality) and harder in contexts without such asymmetry. This is precisely what is observed [31, 34].

11.5 The Tritone as the Devil in the Arnold Tongue

The tritone's unique status in music theory – the only interval actively stigmatised across cultures, forbidden in medieval sacred music, and perceived as maximally dissonant – receives a quantitative explanation. Its Farey rank of 77 makes it the most complex ratio in common musical use, and its position between the two widest non-fundamental tongues (fifth and fourth) places it in the maximally unstable region of parameter space. The cultural response (avoidance, prohibition, association with evil) is the human behavioural signature of an Arnold tongue overlap zone: a region of perceptual instability that produces aversion rather than categorisation.

11.6 Fibonacci Numbers in Scale Structure

The dominance of pentatonic (5-note) and heptatonic (7-note) scales, and the relative rarity of hexatonic (6-note) scales, is consistent with the broader prediction that Fibonacci numbers occupy a privileged position in perceptual organisation [24]. In the Consonance Principle framework, Fibonacci numbers bridge the rational (frequency-locking) and irrational (geometric-packing) stability bands. A 5-note scale samples the octave at intervals that are both consonant (Farey-rational) and maximally non-redundant (close to irrational spacing), producing the optimal balance between harmonic coherence and perceptual distinctness. A 7-note scale adds two more points at the next level of this same trade-off. A 6-note scale, not being a Fibonacci number, does not optimise either criterion and occupies neither stability band.

11.7 Implications for Linguistic Relativity (Sapir-Whorf)

The results bear directly on the Sapir-Whorf hypothesis – the claim that language shapes perception [40, 41]. Two versions of the hypothesis exist:

Strong Sapir-Whorf: Language *determines* perceptual categories. Speakers of different languages perceive the world differently because their languages provide different category structures.

Weak Sapir-Whorf: Language *influences* the ease of perceptual discrimination near category boundaries, without determining the categories themselves.

The Farey hierarchy evidence supports neither version in its standard formulation and instead suggests a third position:

Arnold tongue reframing: Languages do not create perceptual categories. Languages *sample* a pre-existing hierarchy of perceptual attractors imposed by the physics of neural oscillatory coupling. The hierarchy is universal; the sampling depth varies. Languages with more terms sample deeper into the Farey hierarchy, resolving narrower tongues. Languages with fewer terms sample only the widest tongues. But the hierarchy itself – the set of available categories and their relative widths – is determined by physics, not by language.

This reframing preserves the empirical findings that motivated weak Sapir-Whorf (boundary effects near language-specific category edges) while providing a principled explanation for why category boundaries converge cross-linguistically. The boundaries are not arbitrary; they are Arnold tongue edges. Languages that share the same terms share the same tongue boundaries. Languages that differ do so because they sample to different depths of the same underlying hierarchy.

This does not mean language is irrelevant. Language determines *which* narrower tongues are resolved in a given speech community – whether “cyan” gets a name, whether “saudade” is lexicalised, whether “left” and “right” are distinguished. But the underlying perceptual topology – the set of possible categories and their stability ordering – is fixed by physics.

11.8 The Mechanism is Physics, The Manifestation is Culture

It may seem reductive to attribute cross-cultural variation in vocabulary to nonlinear dynamics. But the claim is not that culture is irrelevant. The claim is that culture operates *within* a constraint space defined by physics:

- Physics (Arnold tongue locking) determines the *hierarchy* of possible categories
- Neuroscience (domain-specific salience weighting) determines the *modulation* of the hierarchy
- Culture (communicative need, ecological salience) determines the *depth* to which the hierarchy is sampled
- Language (lexicalisation) determines the *names* assigned to resolved categories

All four levels are necessary to explain any specific language’s vocabulary. But only the first level explains the *universals* – why all languages that have colour terms beyond black and white name red first, why all languages distinguish pleasant from unpleasant emotion, why all cultures produce music with fifths and octaves. The universals are physics. The variation is culture. The framework specifies exactly where the boundary between them lies.

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12. Predictions

The following predictions are falsifiable and, in most cases, testable with existing data or methods.

12.1 Taste Hierarchy Survey

Prediction: A systematic cross-linguistic survey of taste vocabulary, conducted with Berlin-Kay-style controlled elicitation across 50+ languages, will reveal implicational stages analogous to colour: (i) pleasant/unpleasant (universal), (ii) sweet vs. not-sweet, (iii) salty vs. bitter/sour, (iv) bitter vs. sour, (v) umami as a late term. The order will follow Farey complexity within the taste space.

Test: Commission a World Taste Survey using the same methodology as the WCS.

12.2 Olfactory Absence Confirmed

Prediction: Languages with rich abstract olfactory vocabularies (Jahai, Maniq, Seri) will show no internal Farey ordering among their odour terms. The ordering of odour categories, if any exists, will be determined by ecological frequency of exposure, not by position in a ratio space.

Test: Analyse the internal structure of Jahai odour vocabulary for Farey hierarchy. Predict null result.

12.3 Cross-Linguistic Boundary Clustering

Prediction: In every continuous sensory domain, the perceptual boundaries between categories should cluster at Farey fractions of the relevant continuous dimension. Specifically: - Colour: green-blue boundary at 4/3 of the visible octave (~495 nm) across all languages with the distinction - Music: scale-step boundaries at Farey fractions across all cultures - Temperature: warm-cool boundary at the Farey mediant of the hot-cold range across all languages with the distinction

Test: Cross-linguistic boundary location analysis within WCS colour data, DaMuSc scale data, and Koptjevskaja-Tamm temperature data.

12.4 AI Language Models

Prediction: Large language models (LLMs) trained on multilingual corpora will develop internal representations of perceptual categories that mirror the Farey hierarchy. Specifically, the embedding distances between colour terms, emotion terms, and temperature terms in multilingual LLMs should reflect Farey complexity: universal categories should be more separated in embedding space, while “untranslatable” terms should fall between universal categories at positions corresponding to Farey mediants.

Test: Analyse the embedding geometry of perceptual vocabulary in multilingual LLMs (mBERT, XLM-R, GPT-4 embeddings) for Farey structure.

12.5 Novel Perceptual Spaces

Prediction: When humans are exposed to novel continuous perceptual dimensions through technology (AR/VR haptic feedback, direct neural stimulation, synaesthetic cross-modal mappings), they will develop categorical vocabulary for the new dimension in Farey order: binary opposition first, then ternary, then quaternary, with mediants resolved only after flanking categories are established.

Test: Longitudinal study of vocabulary development in VR environments with novel continuous sensory dimensions (e.g., haptic temperature gradients mapped to visual dimensions).

12.6 Developmental Universals

Prediction: Children acquiring any continuous-domain vocabulary should pass through the same Farey stages regardless of language or culture: binary first, then ternary, then quaternary. The age of acquisition for each stage should correlate with Farey complexity across domains (colour, temperature, emotion, space).

Test: Cross-domain developmental study tracking vocabulary acquisition in colour, temperature, emotion, and spatial terms simultaneously in children from 3+ language backgrounds.

12.7 Neural Oscillatory Correlates

Prediction: The perceptual boundaries between categories should correspond to measurable transitions in neural oscillatory dynamics. Specifically, stimuli at Arnold tongue boundaries should elicit increased neural variability (higher entropy in EEG/MEG signals) compared to stimuli at tongue centers, reflecting the perceptual instability at tongue edges.

Test: EEG/MEG study presenting stimuli systematically tiling a continuous perceptual dimension (e.g., colour hue at 5 nm intervals) while measuring oscillatory dynamics. Boundary positions should show entropy peaks at predicted Farey fraction locations.

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13. Conclusion

The Farey hierarchy governs how human languages develop perceptual vocabulary. Across five continuous sensory domains – colour, music, temperature, emotion, and spatial relations – encompassing approximately 3,500 languages, the same pattern recurs: the simplest frequency ratios produce the most universal categories, complexity ordering predicts acquisition sequence, and intermediate fractions in Arnold tongue overlap zones produce systematic gaps. The combined evidence yields $p = 4.75 \times 10^{-11}$. Smell, as a discrete-recognition system without continuous ratio space, shows no such hierarchy – a predicted null that confirms the mechanism requires continuous perceptual input.

This result resolves a 57-year-old puzzle opened by Berlin and Kay's discovery of colour term universals. The answer was never domain-specific. It was never about colour per se, or about the particular physiology of human vision, or about the statistical structure of natural scenes. It was about the mathematics of nonlinear oscillatory coupling applied to any continuous perceptual dimension processed by neural oscillator populations. The mechanism is Arnold tongue locking. The hierarchy is the Farey sequence. The universals are physics.

Three implications stand out.

First, the "untranslatable" experiences that have been cited as evidence for radical linguistic relativity are not evidence that languages create incommensurable perceptual worlds. They are narrow Arnold tongues – perceptual states that are real, discriminable, and universal in the sense that they exist in every human's perceptual parameter space, but whose resolution into named categories requires sampling depth that varies across languages. Saudade, cyan, and the tritone are the same phenomenon in different domains.

Second, the predicted null in olfaction is not a footnote but a structural result. It delineates the boundary of the mechanism with precision: Arnold tongue hierarchy requires continuous perceptual space. Where that prerequisite is met, the hierarchy appears (5/5 domains). Where it is not, the hierarchy is absent (1/1 domains). This is the signature of a genuine physical mechanism, not a flexible analysis framework.

Third, this result connects linguistics, nonlinear dynamics, neuroscience, and cultural evolution through a single mathematical structure. The Farey sequence was discovered by a geologist in 1816. Arnold tongue theory was developed by a mathematician in 1961. Berlin and Kay published their colour universals in 1969. These three bodies of work have existed in separate literatures for decades. The connection between them – that the order of colour term acquisition IS the Farey sequence applied to the visible octave – was invisible because no one had reason to look across all three disciplines simultaneously.

The visible spectrum spans one octave. The auditory system explicitly processes octaves. Temperature perception is a continuous axis. Emotional experience tiles a continuous valence-arousal space. Spatial perception is a continuous angular domain. In every case, the neural systems that process these dimensions are coupled oscillators, and coupled oscillators lock to simple ratios, and simple ratios are named first.

Human languages do not create perceptual categories. They discover them, in the same order, for the same reason, everywhere on Earth. The order is Farey. The reason is Arnold. The phenomenon is universal.

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Data Availability

All data used in this analysis are from published sources cited in the reference list. The World Color Survey data are available at https://www1.icsi.berkeley.edu/wcs/. The DaMuSc database is publicly available. Lexibank data are available at https://lexibank.clld.org/. Jackson et al.'s emotion semantics data are available through the *Science* supplementary materials. Computation code is available from the author upon request.

Competing Interests

The author declares no competing interests. The theoretical framework (Consonance Principle) is the subject of a patent application filed with IP Australia (March 2026) covering computational methods; the empirical analyses presented here are open.

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